

Supplementary Material

Finite-time direction–wavenumber selection in evolving HCP
crystal plasticity:
exact descriptor reduction and path-dependent amplification

1. Complete fixed-candidate norm–selector screen

Figure 1 reports the complete terminal 11-by-7 screen that was previously summarized by representative columns. Each cell gives the winning log gain and abbreviated branch label among the three registered onset- x , onset- y and terminal- y candidates. This figure is a fixed-candidate sensitivity screen; it is not the re-optimized result in Section 2.

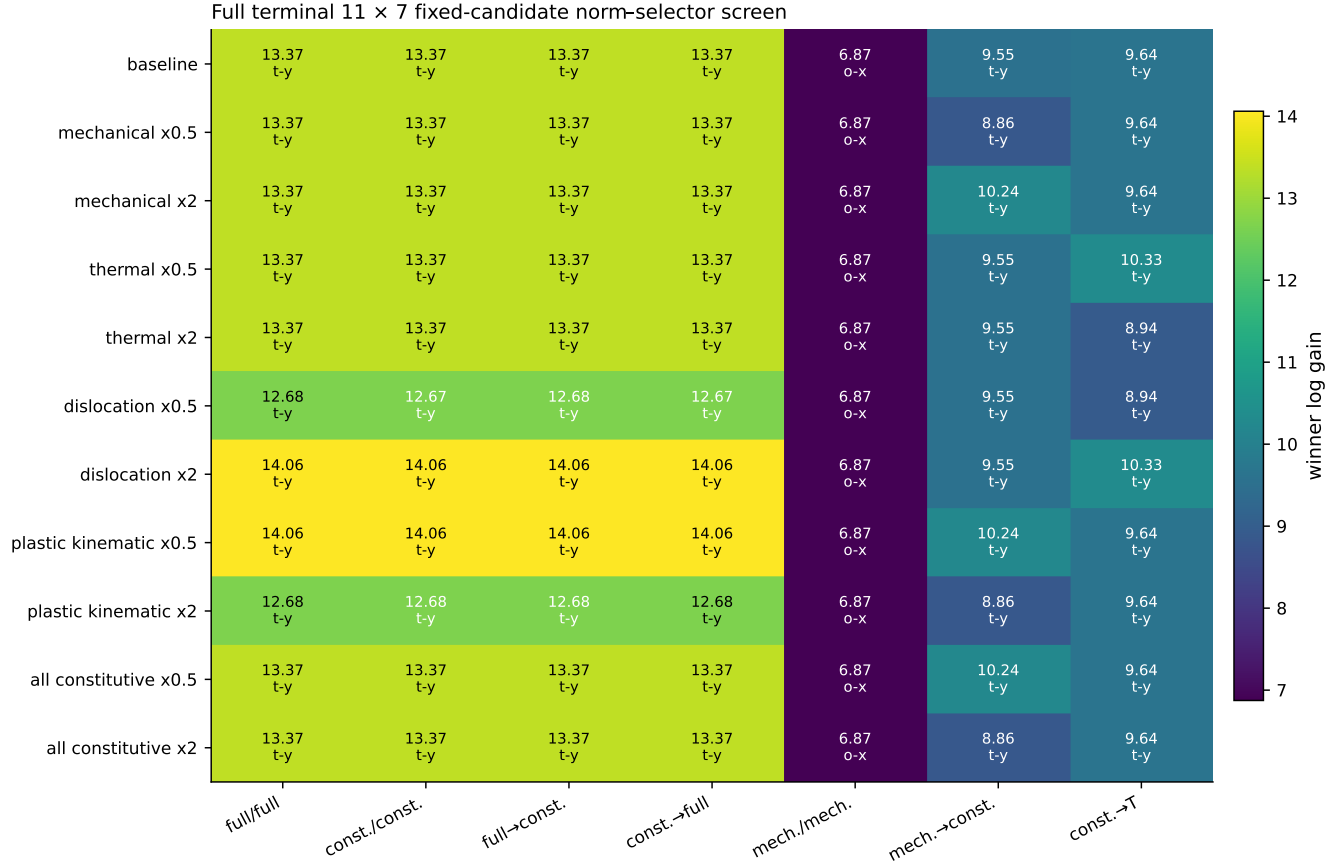


Figure 1: Complete terminal fixed-candidate norm-selector matrix. The rows are the baseline norm and ten factor-two coordinate-scale perturbations; the columns are all seven declared input-output maps. Cell annotations report log gain and the winning registered branch.

2. Re-optimized direction–wavenumber and matched freezing

Direction and wavenumber are re-optimized for all seven selectors under the baseline and four pre-registered adverse norms. The singular-value audit applies the formal 5% relative-gap threshold, and the piecewise-frozen audit uses exactly the same metric and endpoint selectors as the time-ordered map. Machine-readable coordinates, gains, complete singular values/vectors, baseline generators, convergence rows and all optimizer evaluations are supplied in `05_results/ijp_reference_reoptimization_v4.json`, `05_results/ijp_reference_reoptimization_v4_arrays.npz` and `05_results/ijp_reference_reoptimization_v4/`. V3 is retained as the lower-resolution reconnaissance and as the previous-candidate baseline, not as the reference optimization result. The synthesized figure is included in the main paper rather than duplicated here.

Table 1: Complete direct reference-objective re-optimization. Every row is refined on the 129-state/four-substep propagator; $\Delta \log G$ is relative to the V3 retained coordinate reevaluated on that same reference map. Class I is a finite-band interior candidate, LW is a lower-boundary long-wave branch and HF is a high-frequency reversible branch; neither boundary class is interpreted as a localization wavelength.

Horizon	Norm	Input-output	k^* (m ⁻¹)	$\log G$	G	$\Delta \log G$	stat.?	class
onset	all constitutive $\times 2$	constitutive/constitutive	54550.4	0.621942	1.86254	+2.252e-04	yes	I
onset	all constitutive $\times 2$	constitutive $\rightarrow$ full	21315.1	2.969294	19.4782	+3.769e-01	yes	I
onset	all constitutive $\times 2$	constitutive $\rightarrow T$	300	0.000640	1.00064	+6.775e-04	yes	LW
onset	all constitutive $\times 2$	full $\rightarrow$ constitutive	18285.4	0.630114	1.87782	+4.711e-03	yes	I
onset	all constitutive $\times 2$	full/full	283934	9.770385	17507.5	+8.736e-02	yes	HF
onset	all constitutive $\times 2$	mechanical $\rightarrow$ constitutive	18358.7	-1.272480	0.280136	+3.914e-01	yes	I
onset	all constitutive $\times 2$	mechanical/mechanical	283934	9.770385	17507.5	+8.736e-02	yes	HF
onset	baseline	constitutive/constitutive	54550.4	0.621942	1.86254	+2.252e-04	yes	I
onset	baseline	constitutive $\rightarrow$ full	20334.6	2.278484	9.76187	+3.977e-01	yes	I
onset	baseline	constitutive $\rightarrow T$	300	0.000640	1.00064	+6.775e-04	yes	LW
onset	baseline	full $\rightarrow$ constitutive	18322	0.655262	1.92565	+1.792e-02	yes	I
onset	baseline	full/full	283934	9.770385	17507.5	+8.736e-02	yes	HF
onset	baseline	mechanical $\rightarrow$ constitutive	18358.7	-0.579333	0.560272	+3.914e-01	yes	I
onset	baseline	mechanical/mechanical	283934	9.770385	17507.5	+8.736e-02	yes	HF
onset	dislocation $\times 2$	constitutive/constitutive	80968.6	1.093127	2.98359	+2.371e-04	yes	I
onset	dislocation $\times 2$	constitutive $\rightarrow$ full	22306.3	2.288688	9.86199	+3.944e-01	yes	I
onset	dislocation $\times 2$	constitutive $\rightarrow T$	300	0.002454	1.00246	+7.563e-04	yes	LW
onset	dislocation $\times 2$	full $\rightarrow$ constitutive	62926.9	1.107570	3.02699	+7.478e-03	yes	I
onset	dislocation $\times 2$	full/full	283934	9.770385	17507.5	+8.736e-02	yes	HF
onset	dislocation $\times 2$	mechanical $\rightarrow$ constitutive	18358.7	-0.579334	0.560271	+3.914e-01	yes	I
onset	dislocation $\times 2$	mechanical/mechanical	283934	9.770385	17507.5	+8.736e-02	yes	HF
onset	mechanical $\times 0.5$	constitutive/constitutive	54550.4	0.621942	1.86254	+2.252e-04	yes	I
onset	mechanical $\times 0.5$	constitutive $\rightarrow$ full	21315.1	2.969294	19.4782	+3.769e-01	yes	I
onset	mechanical $\times 0.5$	constitutive $\rightarrow T$	300	0.000640	1.00064	+6.775e-04	yes	LW
onset	mechanical $\times 0.5$	full $\rightarrow$ constitutive	18285.4	0.630114	1.87782	+4.711e-03	yes	I
onset	mechanical $\times 0.5$	full/full	283934	9.770385	17507.5	+8.736e-02	yes	HF
onset	mechanical $\times 0.5$	mechanical $\rightarrow$ constitutive	18358.7	-1.272480	0.280136	+3.914e-01	yes	I
onset	mechanical $\times 0.5$	mechanical/mechanical	283934	9.770385	17507.5	+8.736e-02	yes	HF
onset	thermal $\times 0.5$	constitutive/constitutive	59355.1	0.622488	1.86356	+5.519e-04	yes	I
onset	thermal $\times 0.5$	constitutive $\rightarrow$ full	21315.1	2.279999	9.77667	+3.993e-01	yes	I
onset	thermal $\times 0.5$	constitutive $\rightarrow T$	300	0.002455	1.00246	+7.568e-04	yes	LW
onset	thermal $\times 0.5$	full $\rightarrow$ constitutive	18322	0.655886	1.92685	+1.793e-02	yes	I
onset	thermal $\times 0.5$	full/full	283934	9.770385	17507.5	+8.736e-02	yes	HF
onset	thermal $\times 0.5$	mechanical $\rightarrow$ constitutive	18358.7	-0.578156	0.560932	+3.914e-01	yes	I

Horizon	Norm	Input-output	k^* (m^{-1})	$\log G$	G	$\Delta \log G$	stat.?	class
onset	thermal $\times 0.5$	mechanical/mechanical	283934	9.770385	17507.5	+8.736e-02	yes	HF
terminal	all constitutive $\times 2$	constitutive/constitutive	7724.01	13.367382	638823	+5.197e-03	yes	I
terminal	all constitutive $\times 2$	constitutive $\rightarrow$ full	7724.01	13.368115	639291	+0.000e+00	yes	I
terminal	all constitutive $\times 2$	constitutive $\rightarrow T$	7724.01	9.636463	15313.1	+0.000e+00	yes	I
terminal	all constitutive $\times 2$	full $\rightarrow$ constitutive	7724.01	13.367443	638861	+5.347e-03	yes	I
terminal	all constitutive $\times 2$	full/full	7724.01	13.368175	639329	+0.000e+00	yes	I
terminal	all constitutive $\times 2$	mechanical $\rightarrow$ constitutive	6592.74	9.181419	9714.93	+8.916e-01	yes	I
terminal	all constitutive $\times 2$	mechanical/mechanical	300000	9.583252	14519.6	+0.000e+00	yes	HF
terminal	baseline	constitutive/constitutive	7724.01	13.367382	638823	+5.197e-03	yes	I
terminal	baseline	constitutive $\rightarrow$ full	7724.01	13.367566	638940	+5.168e-03	yes	I
terminal	baseline	constitutive $\rightarrow T$	7724.01	9.636463	15313.1	+0.000e+00	yes	I
terminal	baseline	full $\rightarrow$ constitutive	7724.01	13.367623	638977	+5.830e-03	yes	I
terminal	baseline	full/full	7724.01	13.367806	639094	+0.000e+00	yes	I
terminal	baseline	mechanical $\rightarrow$ constitutive	6592.74	9.874566	19429.9	+8.916e-01	yes	I
terminal	baseline	mechanical/mechanical	300000	9.583252	14519.6	+0.000e+00	yes	HF
terminal	dislocation $\times 2$	constitutive/constitutive	7724.01	14.060494	1.2776e+06	+5.315e-03	yes	I
terminal	dislocation $\times 2$	constitutive $\rightarrow$ full	7724.01	14.060677	1.27783e+06	+5.167e-03	yes	I
terminal	dislocation $\times 2$	constitutive $\rightarrow T$	7724.01	10.329573	30625	+0.000e+00	yes	I
terminal	dislocation $\times 2$	full $\rightarrow$ constitutive	7724.01	14.060554	1.27768e+06	+5.428e-03	yes	I
terminal	dislocation $\times 2$	full/full	7724.01	14.060737	1.27791e+06	+0.000e+00	yes	I
terminal	dislocation $\times 2$	mechanical $\rightarrow$ constitutive	6592.74	9.874566	19429.9	+8.916e-01	yes	I
terminal	dislocation $\times 2$	mechanical/mechanical	300000	9.583252	14519.6	+0.000e+00	yes	HF
terminal	mechanical $\times 0.5$	constitutive/constitutive	7724.01	13.367382	638823	+5.197e-03	yes	I
terminal	mechanical $\times 0.5$	constitutive $\rightarrow$ full	7724.01	13.368115	639291	+0.000e+00	yes	I
terminal	mechanical $\times 0.5$	constitutive $\rightarrow T$	7724.01	9.636463	15313.1	+0.000e+00	yes	I
terminal	mechanical $\times 0.5$	full $\rightarrow$ constitutive	7724.01	13.367443	638861	+5.347e-03	yes	I
terminal	mechanical $\times 0.5$	full/full	7724.01	13.368175	639329	+0.000e+00	yes	I
terminal	mechanical $\times 0.5$	mechanical $\rightarrow$ constitutive	6592.74	9.181419	9714.93	+8.916e-01	yes	I
terminal	mechanical $\times 0.5$	mechanical/mechanical	300000	9.583252	14519.6	+0.000e+00	yes	HF
terminal	thermal $\times 0.5$	constitutive/constitutive	7724.01	13.368209	639351	+5.246e-03	yes	I
terminal	thermal $\times 0.5$	constitutive $\rightarrow$ full	7724.01	13.368392	639468	+5.149e-03	yes	I
terminal	thermal $\times 0.5$	constitutive $\rightarrow T$	7724.01	10.329575	30625.1	+0.000e+00	yes	I
terminal	thermal $\times 0.5$	full $\rightarrow$ constitutive	7724.01	13.368450	639505	+5.848e-03	yes	I
terminal	thermal $\times 0.5$	full/full	7724.01	13.368633	639622	+0.000e+00	yes	I
terminal	thermal $\times 0.5$	mechanical $\rightarrow$ constitutive	6592.74	9.875435	19446.8	+8.916e-01	yes	I
terminal	thermal $\times 0.5$	mechanical/mechanical	300000	9.583252	14519.6	+0.000e+00	yes	HF

Table 2: Sampled near-optimal sets for the baseline constitutive/constitutive map. The audit combines an independent scrambled-Sobol challenge with local direction rings and full-domain fixed-direction wavenumber profiles; it is not a continuum global certificate.

Horizon/ ε	members	$k_{\min}$ (m^{-1})	$k_{\max}$ (m^{-1})	max angle (deg)
onset/0.1%	414	1194.32	300000	89.451
onset/0.5%	713	346.12	300000	89.951
onset/1.0%	760	300	300000	89.951
onset/2.0%	760	300	300000	89.951
terminal/0.1%	3	7535.66	7724.01	0.012
terminal/0.5%	5	7535.66	8455.15	0.012
terminal/1.0%	8	6716.16	8455.15	0.012
terminal/2.0%	10	6716.16	9486.83	0.012

3. Loading transfer, local sensitivity and titanium landmarks

The V3 table directory supplies the three added loading cases, the branch-local $\pm 20\%$ sensitivity records and logarithmic derivatives, and the bounded external check against synchronized Grade-II CP-Ti stress/temperature landmarks. The latter is a timing-and-scale check, not a batch-calibrated validation. The figure synthesized from those tables is included in the main paper.

4. Full-grid convergence, branch switching and physical metrics

The V4 full-grid receipt `05_results/ijp_v4_full_fourier_convergence_audit_v1.json` supplies the raw Nyquist negative control, matched no-Nyquist control, two-thirds-dealiased 16/32/64-cell ladder, independent time-step and integration-format checks, and the first-failure equation-variable-wavenumber record. The separate receipt `05_results/ijp_v4_start_branch_metric_audit_v1.json` supplies the initial-time sensitivity, all detected branch transitions, four bounded propagation conventions, and the positive-energy-subspace metric audit. The resolved gain converges, while the discarded nonlinear right-hand-side energy keeps complete harmonic closure open.

The minimum absolute secant event rate is $1.72 \times 10^{13} \text{ Pas}^{-1}$ and the maximum thermally activated threshold-rate jump is $1.97 \times 10^{-13} \text{ s}^{-1}$,

Table 3: Registered active-set events on the factor-64 base history. The category signs give the one-sided branch labels; event times are secant estimates inside the listed checkpoint intervals.

Event time (μ s)	Changed contracts	Category change	Role
0.765638	slip04 yield/rate sign	$0 \rightarrow -1$	activation
0.775926	slip06 yield/rate sign	$0 \rightarrow -1$	activation
1.601157	slip07/slip13 yield/rate sign	$0 \rightarrow (+1, -1)$	activation
2.016486	slip07/slip13 yield/rate sign	$(+1, -1) \rightarrow 0$	deactivation

or 1.97×10^{-17} of the imposed shear rate. The corresponding maximum left/right/secant convention change in terminal gain is 0.0622%.

Table 4: Positive-energy-subspace and nullspace-invariance audit. Relative leakage above 10^{-10} rejects a quotient-space interpretation.

Horizon	rank/nullity	$G_{E \rightarrow E}$	$G_{E \rightarrow O}$	$E \rightarrow E$ leakage	quotient?
Near onset	28/41	1.002	1.697	2.38×10^{-4}	no
Terminal	28/41	117.37	73.16	0.9996	no

5. Reproducibility contracts

The complete direct-reference receipt is `05_results/ijp_reference_reoptimization_v4.json`; the V3 reconnaissance receipt is retained at `05_results/ijp_revision_evidence_v3.json`. The exact literature landmarks and extraction record are under `04_reproducibility/literature_digitization/guo2019_pr1/`. Parameter source keys S2–S5 identify the exact reference record or paper, frozen kernel version, public repository commit, configuration paths, SHA-256 checksums and access date.